

Lec 6-2, Oct 15, 2025.

• Definite integrals (5.3)

- Nonintegrable functions
- Integrable functions
- Computation using Riemann sums
- Properties of definite integrals

• Average value of a function (5.3)

• MVT for definite integrals (5.4)

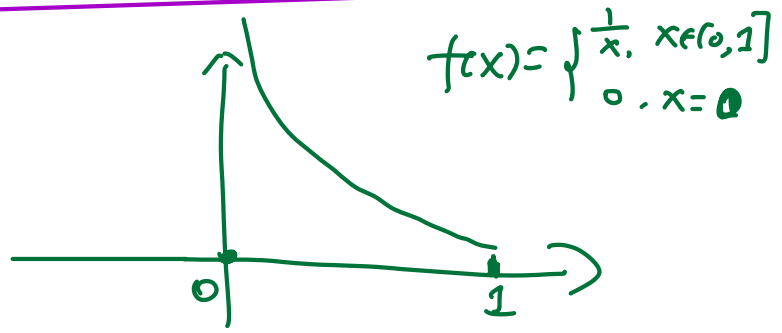
goal:

① Which function is integrable?

② If f is integrable, then how to compute $\int_a^b f(x) dx$?

Q: which function is integrable?

Thm A. If f is unbounded on $[a, b]$, then it is not integrable on $[a, b]$.



Thm B. If f is continuous on $[a, b]$, then f is integrable on $[a, b]$.

Intuition to show this:



① Define $U_P(f)$ and $L_P(f)$ as before, and we are going to show that

$$U_P(f) - L_P(f) \rightarrow 0 \quad \text{as } \|P\| \rightarrow 0.$$

② By continuity, as long as we choose $\|P\|$ to be small enough,

$M_k - m_k$ can be arbitrarily small for any k .

Formally, for any $\varepsilon > 0$, choose $\|P\|$ small enough,

$$\text{s.t. } M_k - m_k = \max_{x \in [x_{k-1}, x_k]} f(x) - \min_{x \in [x_{k-1}, x_k]} f(x)$$

$$< \frac{\varepsilon}{b-a}, \quad \forall k.$$

[This can be done as f is continuous on $[a, b]$.]

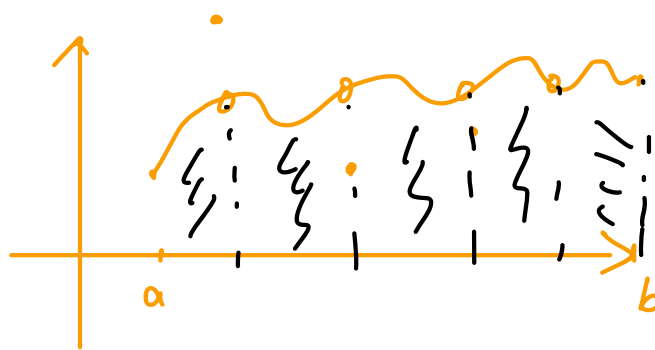
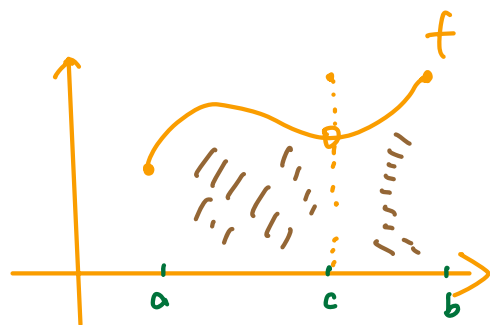
$$\begin{aligned}
 \textcircled{3} \text{ Then } U_P(f) - L_P(f) &= \sum_{k=1}^n M_k \Delta x_k - \sum_{k=1}^n m_k \Delta x_k \\
 &= \sum_{k=1}^n (M_k - m_k) \cdot \Delta x_k \\
 &< \sum_{k=1}^n \frac{\varepsilon}{b-a} \Delta x_k \\
 &= \frac{\varepsilon}{b-a} \sum_{k=1}^n \Delta x_k = \varepsilon \frac{1}{b-a} \cdot (b-a) = \varepsilon.
 \end{aligned}$$

$\textcircled{4}$ Since all Riemann sums $S_P(f) = \sum_{k=1}^n f(c_k) \Delta x_k$ satisfy

$$L_P(f) \leq S_P(f) \leq U_P(f),$$

we know the limit of Riemann sums exists.

Thm C. If f is bounded on $[a, b]$ and is continuous on $[a, b] \setminus X$, where X is a finite set, then f is integrable on $[a, b]$.
 f is cont. $[a, b] \setminus \{c\}$



Intuition to show this :

Δ . a single point does not change the integral on the whole interval !

THEOREM 1—Integrability of Continuous Functions If a function f is continuous over the interval $[a, b]$, or if f has at most finitely many jump discontinuities there, then the definite integral $\int_a^b f(x) dx$ exists and f is integrable over $[a, b]$.

- Important example: bounded but not integrable.

Eg.1. Consider the Dirichlet function

$$f(x) = \begin{cases} 1, & \text{if } x \text{ is a rational number (denoted by } x \in \mathbb{Q}) \\ 0, & \text{if } x \text{ is not a rational number } (x \notin \mathbb{Q}) \end{cases}$$

where the set of rational numbers is

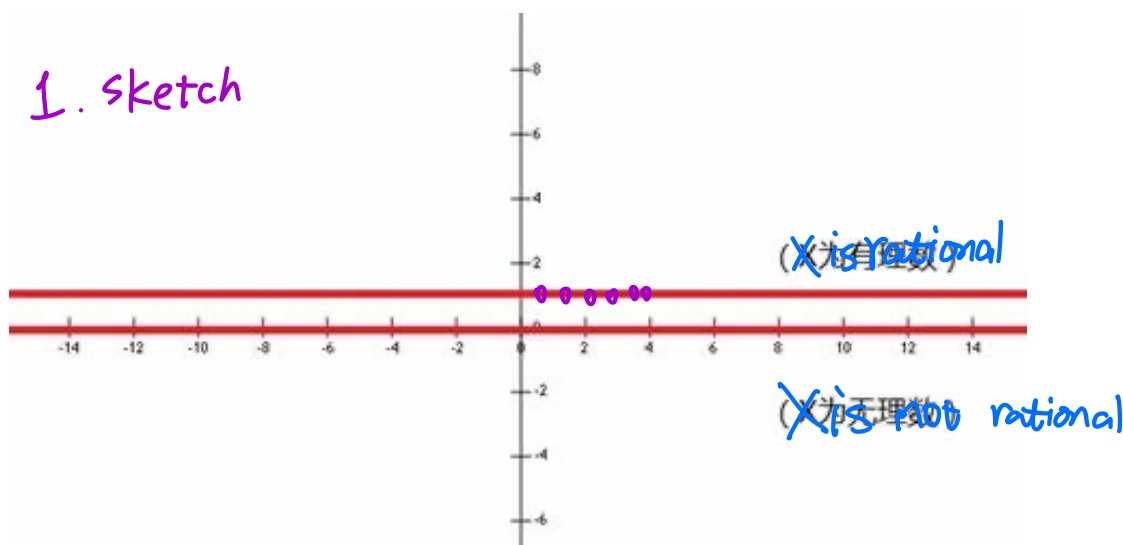
$$\mathbb{Q} = \left\{ \frac{m}{n} : m \in \mathbb{Z}, n \in \mathbb{Z}, n \neq 0 \right\}$$

$$\pi = 3.1415926535 \dots$$

$$\pi - 3$$

and the set of integers is $\mathbb{Z}$.

Properties: 1. sketch



2. f is not continuous at any $x \in \mathbb{R}$. [Optional].

Proof:

$$(i) f \text{ is continuous at } x_0 \Leftrightarrow \lim_{x \rightarrow x_0} f(x) = f(x_0)$$

$$\Leftrightarrow \forall \varepsilon > 0, \exists \delta > 0, \text{ s.t. } \forall x \in D \cap (x_0 - \delta, x_0 + \delta), \\ \text{we have } |f(x) - f(x_0)| < \varepsilon.$$

② if $x_0 \in \mathbb{Q}$, then $f(x_0) = 1$.

But for $\varepsilon = 0.5$, for any $\delta > 0$, there exists some $x_1 \in (x_0 - \delta, x_0 + \delta) \cap D$ and $x_1 \in \mathbb{R} \setminus \mathbb{Q}$, s.t. $f(x_1) = 0$

$$\therefore f(x_1) - f(x_0) = 1 > \varepsilon = 0.5!$$

$\therefore f$ is not continuous at x_0 .

③ if $x_0 \in \mathbb{R} \setminus \mathbb{Q}$, a similar statement holds.

□.

3. f is not integrable on any $[a, b]$.

Intuition:

① any interval contains a rational and an irrational number.

② Hence, for any partition P , in any sub-interval $[x_{k-1}, x_k]$,

we have

$$m_k = \min_{x \in [x_{k-1}, x_k]} f(x) = 0$$

and

$$M_k = \max_{x \in [x_{k-1}, x_k]} f(x) = 1.$$

③ This means

$$U_P(f) = \sum_{k=1}^n M_k \cdot \Delta x_k = \sum_{k=1}^n 1 \cdot \Delta x_k = b-a$$

and

$$L_P(f) = \sum_{k=1}^n m_k \cdot \Delta x_k = \sum_{k=1}^n 0 \cdot \Delta x_k = 0$$

This means the upper sum $= b-a$

and the lower sum $= 0$, no matter how small $\|P\|$ is.

④ As a result, the Riemann Sums do not have a common limit, which means f is not integrable. $\square$.

4. You may think of this as "there is no well-defined area under the graph of $y = f(x)$ ", even f is bounded.

• Computation using Riemann sums

Suppose we know that f is integrable on $[a, b]$. Then we can compute the limit of Riemann sums by choosing any sequence of partitions P such that $\|P\| \rightarrow 0$.

For example, we may choose $a = x_0 < x_1 < x_2 < \dots < x_{n-1} < x_n = b$ where
(P divides $[a, b]$ into n subintervals of equal length) $x_k - x_{k-1} = \frac{b-a}{n}$ ($k=1, \dots, n$)
 $\nwarrow$ $\|P\| = \frac{b-a}{n} \rightarrow 0 \quad (n \rightarrow \infty)$

and take $n \rightarrow \infty$:

$$\int_a^b f(x) dx \quad \xlongequal[\text{if } f \text{ integrable}]{=} \lim_{n \rightarrow \infty} \sum_{k=1}^n f(c_k) \frac{b-a}{n}$$

Here, $c_k \in [x_{k-1}, x_k]$ can be chosen in any way

[You may choose it to make the computation convenient].

Eg. Evaluate $\int_0^1 x^2 dx$ by using Riemann sums.

Sol: Step 1. f is integrable !

As f is continuous on $[0, 1]$, then f is integrable ! ($\because$ Thm B)

$$f(x) = x^2$$

Step 2. Use the equal partition !

we write the Riemann sum with an equal partition:

$$\sum_{k=1}^n f(c_k) \Delta x_k = \sum_{k=1}^n f(c_k) \frac{1-0}{n},$$

where $P = \{x_0, x_1, \dots, x_n\}$ with $x_k = \frac{k}{n}$ and $\underline{c_k} \in [x_{k-1}, x_k]$.

Step 3. Take c_k !

We choose $c_k = x_k$. Then

$$\begin{aligned} \sum_{k=1}^n f(c_k) \Delta x_k &= \sum_{k=1}^n f(c_k) \frac{1-0}{n} \\ &= \sum_{k=1}^n f(x_k) \frac{1-0}{n} \\ &= \sum_{k=1}^n \left(\frac{k}{n}\right)^2 \cdot \frac{1}{n} \\ &= \frac{1}{n^3} \sum_{k=1}^n k^2 = \frac{1}{n^3} \cdot \frac{1}{6} n(n+1)(2n+1). \end{aligned}$$

Step 4 : Take the limit !

Letting $n \rightarrow \infty$, we have

$$\begin{aligned} \lim_{n \rightarrow \infty} \sum_{k=1}^n f(c_k) \Delta x_k &= \lim_{n \rightarrow \infty} \frac{1}{n^3} \frac{1}{6} n(n+1)(2n+1) \\ &= \frac{1}{3}. \end{aligned}$$

As f is integrable, we have

$$\int_0^1 f(x) dx = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(c_k) \Delta x_k = \frac{1}{3}.$$



In Step 3, if we take the left endpoint, i.e., $c_k = x_{k-1}$, we have

$$\begin{aligned} \sum_{k=1}^n f(c_k) \Delta x_k &= \sum_{k=1}^n f(c_k) \frac{1-0}{n} \\ &= \sum_{k=1}^n f(x_{k-1}) \frac{1-0}{n} \\ &= \sum_{k=1}^n \left(\frac{k-1}{n} \right)^2 \frac{1}{n} \\ &= \frac{1}{n^3} \sum_{k=1}^n (k-1)^2 \\ &= \frac{1}{n^3} \sum_{k=1}^{n-1} k^2 \\ &= \frac{1}{n^3} \frac{1}{6} (n-1)n(2n-1). \end{aligned}$$

$$\lim_{n \rightarrow \infty} \frac{1}{n^3} \frac{1}{6} (n-1) n (2n-1) = \frac{1}{6} \lim_{n \rightarrow \infty} \frac{(n^2-n)(2n-1)}{n^3} = \frac{1}{3}.$$

$$\therefore \int_0^1 x^2 dx = \lim_{n \rightarrow \infty} \frac{1}{n^3} \frac{1}{6} (n-1) n (2n-1) = \frac{1}{3},$$

the same result!

□

Properties of definite integrals.

$[a, b]$

Def: For $a < b$, we define $\int_b^a f(x) dx \triangleq -\int_a^b f(x) dx$,

$$\int_a^b f(x) dx \quad \int_a^a f(x) dx \triangleq 0.$$

The intuition behind the first definition is that all Δx_k 's in a Riemann sum become a negative of the original one when x goes from b to a .

TABLE 5.6 Rules satisfied by definite integrals

1. Order of Integration:	$\int_b^a f(x) dx = -\int_a^b f(x) dx$	A definition
2. Zero Width Interval:	$\int_a^a f(x) dx = 0$	A definition when $f(a)$ exists
3. Constant Multiple:	$\int_a^b kf(x) dx = k \int_a^b f(x) dx$	Any constant k
4. Sum and Difference:	$\int_a^b (f(x) \pm g(x)) dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx$	
5. Additivity:	$\int_a^b f(x) dx + \int_b^c f(x) dx = \int_a^c f(x) dx$	
6. Max-Min Inequality:	If f has maximum value $\max f$ and minimum value $\min f$ on $[a, b]$, then	
	$\min f \cdot (b - a) \leq \int_a^b f(x) dx \leq \max f \cdot (b - a).$	
7. Domination:	$f(x) \geq g(x) \text{ on } [a, b] \Rightarrow \int_a^b f(x) dx \geq \int_a^b g(x) dx$	
	$f(x) \geq 0 \text{ on } [a, b] \Rightarrow \int_a^b f(x) dx \geq 0$	(Special case)

linearity:

$$\int_a^b (\alpha f(x) \pm \beta g(x)) dx$$

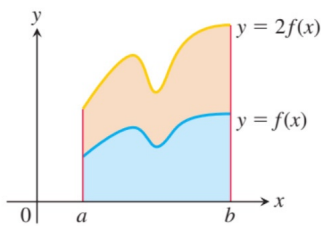
$$= \alpha \int_a^b f(x) dx \pm \beta \int_a^b g(x) dx$$

$$\sum_{k=1}^n (\alpha f(k) \pm \beta g(k))$$

$$= \alpha \sum_{k=1}^n f(k) \pm \beta \sum_{k=1}^n g(k)$$

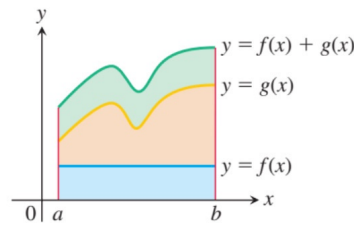
All properties above hold whenever the integrals exist : f could be negative or discontinuous .

• Intuition of properties 3 to 7, in terms of area:



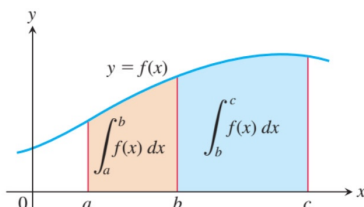
3) Constant Multiple: ($k = 2$)

$$\int_a^b kf(x) dx = k \int_a^b f(x) dx$$



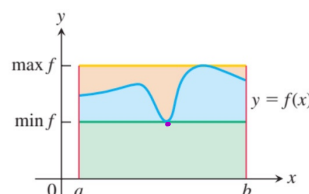
4) Sum: (areas add)

$$\int_a^b (f(x) + g(x)) dx = \int_a^b f(x) dx + \int_a^b g(x) dx$$



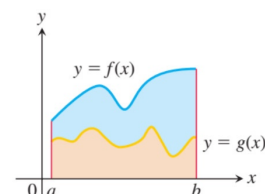
5) Additivity for Definite Integrals:

$$\int_a^b f(x) dx + \int_b^c f(x) dx = \int_a^c f(x) dx$$



6) Max-Min Inequality:

$$\min f \cdot (b - a) \leq \int_a^b f(x) dx \leq \max f \cdot (b - a)$$



7) Domination:

$$f(x) \geq g(x) \text{ on } [a, b] \Rightarrow \int_a^b f(x) dx \geq \int_a^b g(x) dx$$

· For additivity $\int_a^b + \int_b^c = \int_a^c$, b does not need to be in $[a, c]$.

e.g. $\int_3^6 f(x) dx + \int_6^5 f(x) dx = \int_3^5 + \int_5^6 + \int_6^5 = \int_3^5 + (\int_5^6 + \int_6^5) = \int_3^5 + 0$

still holds !

e.g.3. Prove that $0 \leq \int_a^b \sqrt{1+\cos x} dx \leq \sqrt{2} (b-a)$, if $b \geq a$.

Pf: $\because \sqrt{1-1} \leq \sqrt{1+\cos x} \leq \sqrt{1+1}$, $\forall x \in \mathbb{R}$ [when you see $\sin x$, $\cos x$, you should always try first: $-1 \leq \sin x \leq 1$.]

$\therefore$ By max-min inequality, we have

$$0 = 0 \cdot (b-a) \leq \int_a^b \sqrt{1+\cos x} dx \leq \sqrt{2} (b-a).$$

□.

Finite sums

The summation symbol (Greek letter sigma) $\sum$ — a_k is a formula for the k th term.

The index k starts at $k = 1$.

The index k ends at $k = n$.

Algebra Rules for Finite Sums

- Sum Rule:** $\sum_{k=1}^n (a_k + b_k) = \sum_{k=1}^n a_k + \sum_{k=1}^n b_k$
- Difference Rule:** $\sum_{k=1}^n (a_k - b_k) = \sum_{k=1}^n a_k - \sum_{k=1}^n b_k$
- Constant Multiple Rule:** $\sum_{k=1}^n c a_k = c \cdot \sum_{k=1}^n a_k$ (Any number c)
- Constant Value Rule:** $\sum_{k=1}^n c = n \cdot c$ (c is any constant value.)

Definite integral :

The function is the integrand.

x is the variable of integration.

Upper limit of integration b

Integral sign $\int$ *Sum*

Lower limit of integration a

$f(x) dx$

When you find the value of the integral, you have evaluated the integral.

Integral of f from a to b

TABLE 5.6 Rules satisfied by definite integrals

- Order of Integration:** $\int_b^a f(x) dx = - \int_a^b f(x) dx$ A definition
- Zero Width Interval:** $\int_a^a f(x) dx = 0$ A definition when $f(a)$ exists
- Constant Multiple:** $\int_a^b k f(x) dx = k \int_a^b f(x) dx$ Any constant k
- Sum and Difference:** $\int_a^b (f(x) \pm g(x)) dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx$
- Additivity:** $\int_a^b f(x) dx + \int_b^c f(x) dx = \int_a^c f(x) dx$
- Max-Min Inequality:** If f has maximum value $\max f$ and minimum value $\min f$ on $[a, b]$, then

$$\min f \cdot (b - a) \leq \int_a^b f(x) dx \leq \max f \cdot (b - a).$$
- Domination:**

$$f(x) \geq g(x) \text{ on } [a, b] \Rightarrow \int_a^b f(x) dx \geq \int_a^b g(x) dx$$

$$f(x) \geq 0 \text{ on } [a, b] \Rightarrow \int_a^b f(x) dx \geq 0 \text{ (Special case)}$$

Logic

Step 1.

Riemann sums (a lot!): $\sum_{k=1}^n f(c_k) \cdot \Delta x_k$

Step 2.

Two special sums: upper sum and lower sum

$$U_P(f) = \sum_{k=1}^n M_k \cdot \Delta x_k, \quad L_P(f) = \sum_{k=1}^n m_k \Delta x_k.$$

Step 3.

If the above two has a same limit as $\|P\| \rightarrow 0$, we say f is integrable and denote the limit by $\int_a^b f(x) dx$.

$$U_P(f) - L_P(f) \rightarrow 0 \quad (\|P\| \rightarrow 0).$$

$$\int_a^b f(x) dx = \lim_{\|P\| \rightarrow 0} \sum_{k=1}^n f(c_k) \Delta x_k.$$

Step 4.

We find:

- ① f continuous or at most jump finite times $\Rightarrow f$ integrable
- ② f unbounded $\Rightarrow f$ not integrable.

integrable functions



all the functions f on $[a, b]$.

Cont' func'

jump finite times

unbounded

Step 5.

Compute a representative limit for integrable function

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(c_k) \frac{b-a}{n} \quad \text{if } f \text{ is integrable}$$

[we may choose an equal partition and all left endpoints for c_k 's, e.g.]

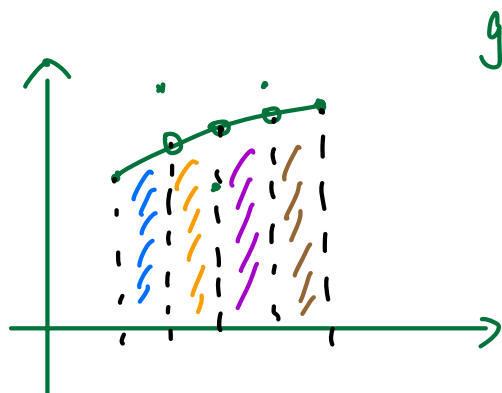
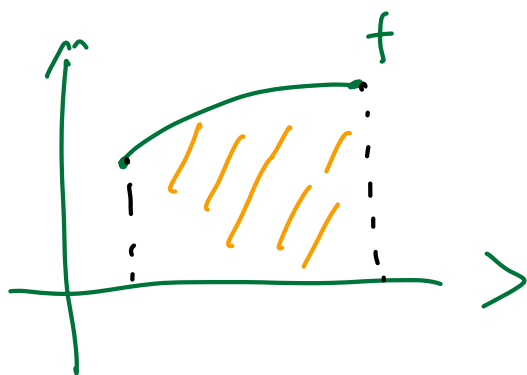
Step 6.

Integration rules for computation.

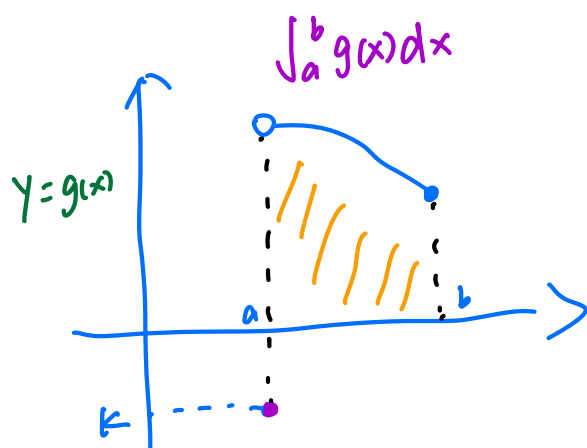
The following is another property of definite integrals.

Thm. If f is integrable on $[a, b]$ and g is the same as f on $[a, b]$, except at finitely many points $x_1, \dots, x_m$, then g is integrable on $[a, b]$, and $\int_a^b g(x) dx = \int_a^b f(x) dx$.

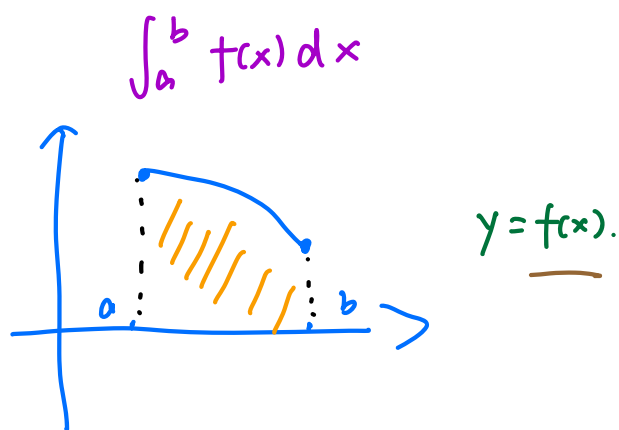
Intuition:



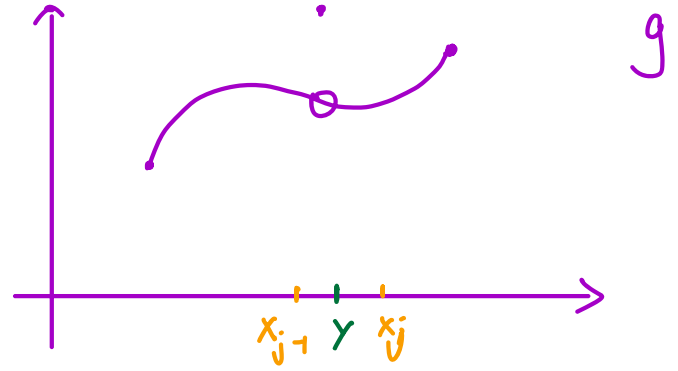
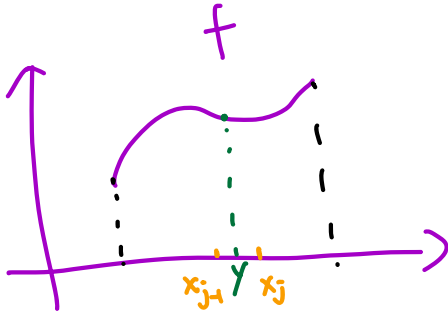
e.g. $\underbrace{g(x)} = \begin{cases} f(x), & \text{if } x \in [a, b] \\ k, & \text{if } x = a \end{cases}$



=



Further intuition.



$$f(y) \neq g(y)$$

• Riemann sum for f :

$$\sum_{k=1}^n f(c_k) \Delta x_k = \sum_{k=1}^{j-1} f(c_k) \Delta x_k + f(c_j)(x_j - x_{j-1}) + \sum_{k=j+1}^n f(c_k) \Delta x_k.$$

• Riemann sum for g :

$$\sum_{k=1}^n g(c_k) \Delta x_k = \sum_{k=1}^{j-1} f(c_k) \Delta x_k + g(c_j)(x_j - x_{j-1}) + \sum_{k=j+1}^n f(c_k) \Delta x_k.$$

• f and g are only different at $y \in [x_{j-1}, x_j]$ and fixed!

• But the subinterval length is going to 0!

• And the limit of all the Riemann sums of f exists!

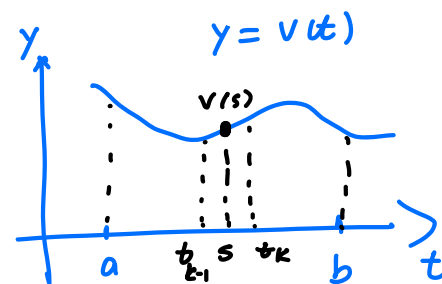
$\Rightarrow$ the limit of all the Riemann sums of g exists!

$\Rightarrow g$ is integrable on $[a, b]$ and $\int_a^b g(x) dx = \int_a^b f(x) dx$

□

Average values / means

① Average speed : $v(t)$: velocity at time t

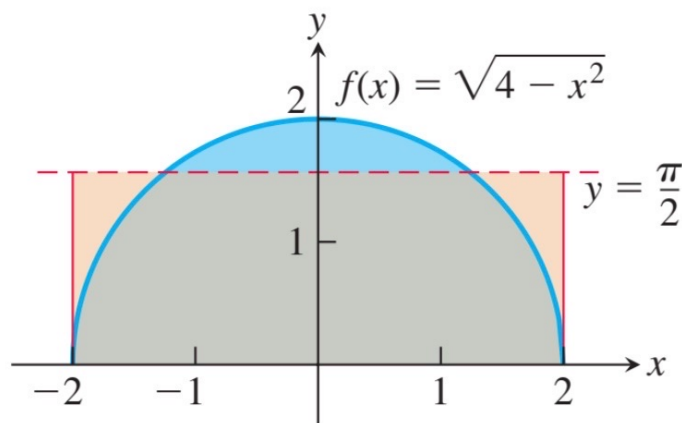


• $\sum_{k=1}^n v(s_k) \Delta t_k \approx$ total displacement travelled from $t=a$ to $t=b$.

• $\int_a^b v(t) dt$ = exact displacement travelled from $t=a$ to $t=b$.

• $\frac{1}{b-a} \int_a^b v(t) dt$ = average velocity over $[a, b]$.

② Average height



• $\int_{-2}^2 \sqrt{4-x^2} dx$ = exact area of f over $[-2, 2]$.

• $\frac{1}{2-(-2)} \int_{-2}^2 \sqrt{4-x^2} dx$ = average height over $[-2, 2]$

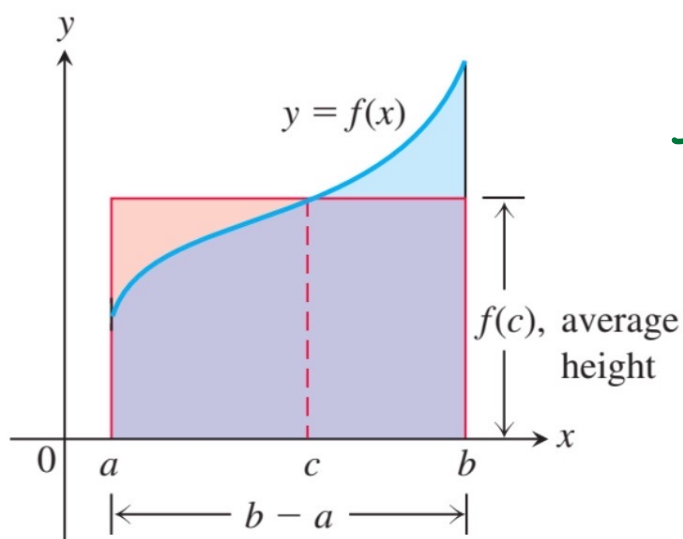
③ In general,

DEFINITION If f is integrable on $[a, b]$, then its average value on $[a, b]$, also called its mean, is

$$\text{av}(f) = \frac{1}{b-a} \int_a^b f(x) dx.$$

Rmk : f may not be nonnegative in the definition above.

• Mean Value Theorem for Definite Integrals



If f is continuous on $[a, b]$,
then there exists some $c \in [a, b]$
with $f(c)$ achieving the average
height of f on $[a, b]$, i.e.,

$$f(c) = \frac{1}{b-a} \int_a^b f(x) dx.$$

5.4.3

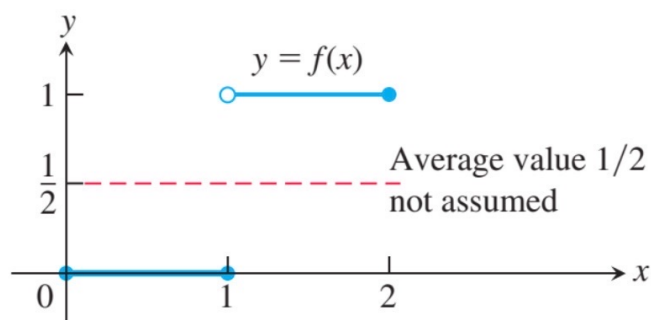
THEOREM 3—The Mean Value Theorem for Definite Integrals

If f is continu-

ous on $[a, b]$, then at some point c in $[a, b]$,

$$f(c) = \frac{1}{b-a} \int_a^b f(x) dx.$$

• Continuity conditions cannot be skipped!



$$f(x) = \begin{cases} 0, & x \in [0, 1]; \\ 1, & x \in (1, 2]. \end{cases}$$

$$\frac{1}{2-0} \int_0^2 f(x) dx = \frac{1}{2}, \text{ but } f(c) \neq \frac{1}{2}, \forall c \in [0, 2]!$$

$$\left(= \frac{1}{2} (\int_0^1 0 dx + \int_1^2 1 dx) = \frac{1}{2} \cdot (1 \cdot 0 + 1 \cdot (2-1)) = \frac{1}{2} (0+1) \right)$$

Pf: Step 0.

If f is a **constant** on $[a, b]$, the statement trivially holds.

Step 1. Analyze! By the **max-min** inequality in integration rules,

the average value $\frac{1}{b-a} \int_a^b f(x) dx$ is between

$$m \triangleq \min_{x \in [a, b]} f(x) \text{ and } M \triangleq \max_{x \in [a, b]} f(x),$$

where there exist $x_m \in [a, b]$ and $x_M \in [a, b]$, s.t.

$$f(x_m) = m \quad \text{and} \quad f(x_M) = M.$$

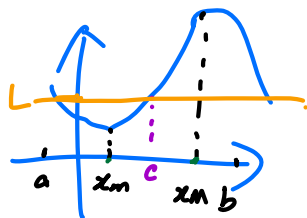
It suffices to show that there exists c between x_m and x_M

$$\text{s.t. } f(c) = \frac{1}{b-a} \int_a^b f(x) dx.$$

$$L = \frac{1}{b-a} \int_a^b f(x) dx$$

Step 2. Use intermediate value theorem (IVT)!

As f is **continuous** and $\frac{1}{b-a} \int_a^b f(x) dx \in [m, M]$,



there exists some c between x_m and x_M s.t.

$$f(c) = \frac{1}{b-a} \int_a^b f(x) dx.$$

□.